

RAJESH PONNAGANTI

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SUMMARY

AI and Machine Learning enthusiast, currently pursuing an MSc degree in Artificial Intelligence and Robotics, with significant experience in building AI-powered applications, machine learning models, and intelligent systems. Skilled in AI and Machine Learning, with expertise in building AI-powered applications, deep learning, large language models, and AI system deployment. Proficient in building end-to-end AI solutions, including computer vision, natural language processing, and autonomous agents, with a strong commitment to building AI systems, including AI assistants, robotics vision systems, and intelligent automation tools.

EDUCATION

Masters, Artificial Intelligence and Robotics 2025 - Present
University of Hertfordshire, Hertfordshire, United Kingdom

B.Tech, Computer Science and Engineering 2021 - Present
KL University, vijayawada, India 8.86 CGPA

TECHNICAL SKILLS

Programming: Python, Java

AI / Machine Learning: TensorFlow, PyTorch, Scikit-learn, OpenCV, Deep Learning, NLP, Computer Vision

AI Systems: LLM Applications, Agentic AI Systems, Retrieval Augmented Generation (RAG)

Web & Deployment: Flask, Django, Streamlit, FastAPI

Databases: MySQL, MongoDB, Vector Databases

Tools: Git, Docker, Linux

PROJECTS

Voice AI Assistant for Raspberry Pi 2025
Developed an edge-deployed voice-controlled AI assistant capable of understanding natural language commands and executing tasks using speech interfaces and LLM-based reasoning.

- Implemented a speech pipeline integrating **speech-to-text**, **natural language understanding**, and **text-to-speech**, combined with **LLM-based reasoning** to interpret user commands and automate tasks such as information retrieval and system control.
- Designed a modular architecture optimized for **edge deployment on Raspberry Pi**.

Autonomous AI Research Assistant 2025
Built an autonomous AI system that researches technical topics, retrieves academic papers, and generates structured summaries using LLM-powered multi-agent workflows.

- Developed a **multi-agent AI pipeline** capable of searching, retrieving, and summarizing research papers.
- Implemented intent classification and task execution modules to enable the assistant to perform automated actions such as information retrieval, system commands, and API interactions.
- Implemented a **Retrieval-Augmented Generation (RAG)** architecture using vector embeddings and semantic search to automate document ingestion, text extraction, and summarization of academic PDFs.

PAPERS PUBLISHED

AN EMPOWERING SAFER ONLINE SPACES: ADVANCEMENTS IN COMPREHENSIVE CYBERBULLYING DETECTION THROUGH NLP AND MACHINE LEARNING ALGORITHMS Aug 2024

- Implemented advanced NLP and machine learning algorithms to develop a comprehensive cyberbullying detection system, promoting safer online spaces.

ACTIVITIES

School of Competitive Coding (SOCC) Club July 2023 - May 2024
Led the SOCC Club, boosting its profile and membership while organizing coding competitions, hackathons, and workshops.

- Successfully led the SOCC for one year, enhancing the club's profile and membership.
- Organized and conducted various technical events, including coding competitions, hackathons, and workshops